EMRE AVŞAR

CONTROL SYSTEMS & SOFTWARE ENGINEER

İstanbul, Turkiye | +90 (554) 251 24 65 | avsaremre94@gmail.com | linkedin.com/in/eavsar | emreavsar94.github.io

Control Systems & Software Engineer with industrial R&D experience in safety-critical systems. Specialized in Model-Based Design (MBD) using MATLAB & Simulink, real-time control algorithm development, and end-to-end V-cycle execution. Proven track record in developing VCU features, AUTOSAR architecture, and autonomous driving functions at Togg. Expert in bridging advanced control theory with embedded target deployment.

EDUCATION

Ph.D. in Control & Automation Eng.
İstanbul Technical University
2026 – Present

M.Sc. in Mechatronics Eng.
İstanbul Technical University
2017 – 2021 | GPA: 3.13

B.Sc. in Control & Automation Eng.
Yıldız Technical University
2012 – 2017 | GPA: 3.05

Beşiktaş Atatürk High School
Science and Mathematics
2008 – 2012

CORE SKILLS

CORE EXPERTISE
Model-Based Design (MBD), Control System Design, State Estimation & Filtering, System Dynamics, Sensor Fusion, V-Cycle Management.

SOFTWARE & SIMULATION
MATLAB / Simulink, Stateflow, Simulink Coder, AUTOSAR Toolchains.

PROGRAMMING
C/C++, Python, Scripting (MATLAB/Python), C#.

TESTING & CALIBRATION
ETAS INCA, Vector CANape / CANoe, MDA, Concerto, MIL/SIL/HIL Testing.

INDUSTRIAL AUTOMATION & ROBOTICS
Real-Time Motion Control, EtherCAT, PLC, SCADA, HMI Architecture.

REFERENCES

Dr. Volkan ARAN
Motor Control Engineer / Powertrain SW & Cal Substream Leader, Togg

Prof. Dr. Tufan KUMBASAR
PhD Supervisor, İstanbul Technical University

PROFESSIONAL EXPERIENCE

TOGG | *E-Powertrain Control Systems & SW Engineer* *10.2022 – Present*

- **Model-Based Design:** Modeling and simulating high-fidelity systems using MATLAB & Simulink; utilizing advanced scripting for model optimization and automated data post-processing.
- **Feature Ownership & Integration:** Leading the end-to-end development of critical VCU features (including Range Estimation, LV Management, Gear Shifting Logic) encompassing requirement management, control system design, and software integration/verification. Spearheading high-level system integration, state-machine logic design, and on-vehicle validation of autonomous driving and powertrain functions.
- **AUTOSAR & Embedded Architecture:** Designing and developing Model-Based Application Software (ASW) components aligned with AUTOSAR standards based on complex system requirements.
- **V-Cycle & Testing:** Executing MIL/SIL/HIL V-cycle validation, testing, and calibration using ETAS INCA, Vector toolchains, MDA, and Concerto across relevant environments.

ALTINAY ROBOT TECHNOLOGIES | *Control and Automation Engineer* *09.2021 – 09.2022*

- **Real-Time Motion Control:** Developed and implemented real-time motion control algorithms using EtherCAT protocols for industrial PC-based architectures.
- **Control Design & Simulation:** Designed and integrated test/simulation environments for a TÜBİTAK-funded robotics project, successfully implementing and validating complex control algorithms.
- **HMI & Diagnostics Optimization:** Formulated intuitive HMI/User Interfaces to optimize real-time system monitoring and efficiency.

Internships: Dal Electric (2017) | KABMAK Eng (2016) | Kognitron IT (2015)

RESEARCH & THESIS PROJECTS

Fault Tolerant Multi-Robot Targeting | *M.Sc. Thesis* | *ITU* | *MATLAB Simulation*
Developed autonomous route planning laws and fallback algorithms for multi-agent platforms under fault scenarios within integrated simulation environments.

Optimal Path Planning in Mobile Robots | *Graduation Project* | *YTU* | *MATLAB, Arduino*
Designed shortest-path algorithms in Simulink and integrated vision-based target tracking loops via MATLAB image processing.

Dynamic Threat Tracking System | *Design Project* | *YTU* | *C++ & OpenCV, Servos*
Implemented real-time visual tracking loops using C++ & OpenCV, closed with microcontroller-driven servo targeting mechanisms.

ACADEMIC PUBLICATIONS

- **E. Avşar**, F. Çalışkan, "Bilinmeyen Ortamda Çoklu Robotlarla Arızaya Toleranslı Hedefe Ulaşma," *TOK*, 2021.
- **E. Avşar**, F. Çalışkan, "Comparison of Algorithms for Targeting with Multiple Robots in Unknown Environment," *ISASTECH*, 2021.
- C. Zoroğlu, **E. Avşar**, F. Oynak, B. Erol, A. Delibaşı, "Mobil Robotlarda Engelden Kaçınma Algoritmaları ve Optimal Rota Planlaması," *TOK*, 2017.